

PLAN

$$\min_b \mathbb{E}[\text{loss}] \text{ s. t. } \bar{b}_t \leq B$$

WHAT IT DOES

Hybrid gradient $g(\alpha) = \alpha g_{\text{pw}} + (1 - \alpha) g_{\text{LR}}$, with $\alpha^* = (V_L - \text{Cov}) / (B^2 + V_p + V_L - 2 \text{Cov})$.

A PID dual $\lambda_k = [K_p e_k + K_i I_k + K_d \max(0, e_k - e_{k-1})]_+$ driven by measured constraint error.

At decision time, projected gradient through the rollout with exact projection onto the budget.

WHAT HELD UP

12.0 points over the operational heuristic on 1500 held out fires, paired t of 11.15. The gap widens to 13.6 with the hand written physics removed. Violations 7.3 percent to 0.

WHAT DID NOT

Equation 8 is degenerate: the measured bias squared is 0.0013, so it and the variance only rule pick the same alpha. The amortised policy reaches only 10.4 percent, the same failure as safe RL.

The mechanism is measured, not asserted. At a one day horizon the planner and the heuristic are indistinguishable, t of 1.40, exactly as theory requires since that objective is separable over cells. The gap reaches 14.3 points at five days, and the heuristic degrades because it keeps optimising for today.